Executive PG Diploma - Machine Learning

End-Semester Exam **29-Nov-2025** **Max Marks:**

Instructions: Handwritten and Printed slides/notes permitted.

1. Indicate whether the following statements are true or false. Justify your answer in a line or two. [1×5 = 5]
   a. In support vector classifiers, the weight vector is always parallel to the vector corresponding to feature elements of the support vector (line joining origin to the support vector $\mathbf{x}_n$).
   False. The weight vector is normal to the hyperplane, and will thus connect to support vectors in the normal direction. Line joining origin and the support vector can be in any direction.

   b. A Lasso regression model effectively yields a modified, lower dimensional feature space.
   True: Lasso model can effectively reduce some of the coefficients (weights) to zero.

   c. Determining age of a pet dog using the database of lifespan of the same breed is a good problem for reinforcement learning.
   False: This is a simple regression problem or statistical average.

   d. K-nearest neighbor method with lower value of K always generalizes well compared to logistic regression.
   False: This may happen for a smaller dimensional data set, but not in general.

   e. You are tasked to build a model and estimate the out of sample error from 10,000 data points. Let us say you have used 8,000 data points for building the model. A cross validation data set of size 200 is expected to give better estimate of E_{out} compared to the one with 100 data points.
   False: As the number of data points in the CV set reduces, the E_{cv} approaches E_{out}

2. Consider a data $[y_1, y_2, \ldots, y_M]$ of M points, for which a representative value is to be found. This is equivalent to finding a regression fit ($y = c$) that is independent of any feature. Suggest a way to evaluate the same, for the following two cases: [[redacted]]
   [4]
   a. Data without any outliers
   Simple regression of $y = c$ will yield $c = \frac{1}{N}\Sigma_{i=1}^{N}(y_i)$

   b. Data with some outliers.
   If data has outliers, the mean can be strongly affected by the same and may not be representative. Hence a better choice would be to use the median.

3. An electric car company collected data of simulated tests (10,000 in number) and observed that the car crashed with a pedestrian about 100 times in various operating conditions. A classification model is to be built to relate features with crash or otherwise. Propose a weighted classification error metric for the problem and briefly justify. [[redacted]]
   [5]

$$E_{in} = \frac{1}{\mathrm{N}}\sum \begin{Bmatrix} 1 & if\ y = +1 \\ 100 & if\ y = -1 \end{Bmatrix} \cdot [\![y \neq h(\mathbf{x})]\!]$$

Where
Y = +1: Crash. E=1 (false positive) meaning lower penalty for detecting crash when it should not be crashing (equivalent to detecting a regular employee as an intruder). False positives are somewhat acceptable.
Y = -1: No Crash (False negative) E=100 can be a higher penalty for classification model detecting no crash (continue to drive forward) thinking that there is no danger.

100 is preferred so that 100*100 = 10000, which is a number close to the data of no crash. This will make the data-sets more balanced for classification.

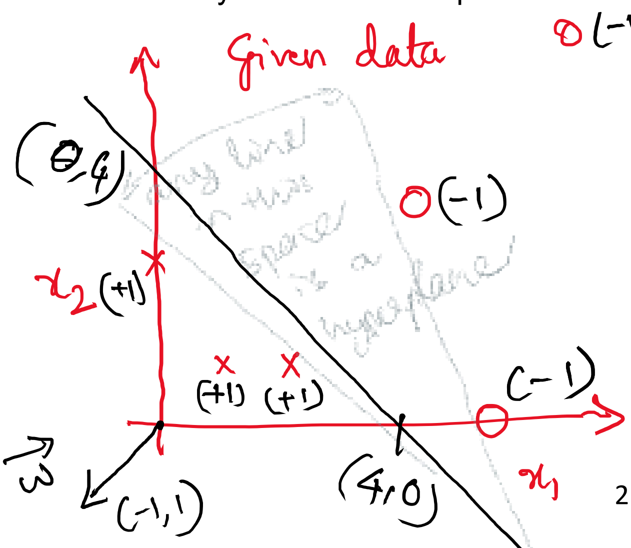
4. A data analyst builds the following regression models from N data points – (i) KNN Regression (ii) Linear Regression (iii) Support vector regression and (iv) Decision Tree Regression. A single outlier was added to the to the dataset that has a high leverage (it's far from the other data points on the x-axis) but is also far from the original regression line, and model was re-built. Arrange the 4 methods in the order of decreasing influence of the outlier. Justify your answer. [[redacted]]
5

(i) Since the outliers are far from other data points on x-axis KNN regression would remain the same for most of the original range of features. However, for $x = x_{outlier}$, the prediction is affected by the outlier. RANK 2 or 3
(ii) Linear regression: The entire regression curve slightly shifts as the MSE is changed. RANK 1
(iii) SVR: Absolutely no effect of outlier as support vectors are only those within the ϵ band. RANK 4
(iv) Decision tree regression prediction is affected only where the basket is near the outlier. RANK 2 or 3

5. Consider the data given below and answer the three questions. [redacted] [5 + 5 + 5]

Feature	X_1	1	0	6	5	2	7
Feature	X_2	1	3	0	4	1	9
Label	Y	1	1	-1	-1	1	-1

a. Using a 2D visual representation, obtain an equation of any separating hyperplane that correctly classifies these points. Show the weight vector.



Consider the black line passing through (4,0) & (0,4)
(Other lines also possible such as $x_1 = 4$!)

$x_1 + x_2 = 4 \Rightarrow w_1x_1 + w_2x_2 + w_0 = 0$

$w_1 = 1,\ w_2 = 1,\ w_0 = -4$

check

Is this a good hypothesis? For $(1,1)$: $1+1-4=-2$

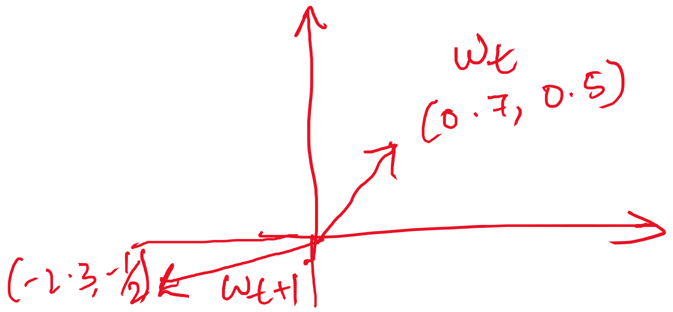
$\text{sign}(-2) = -1$ ✗

Correct hypothesis: $h = \text{sign}(-x - y + 4) \Rightarrow w = \begin{bmatrix} 4 \\ -1 \\ -1 \end{bmatrix}$. weight vector

b. Given the guesses: $w_1 = 0.7$, $w_2 = 0.5$, and $w_0 = -1$, perform calculation of one update iteration on any one data point using the perceptron learning algorithm, and show the two weight vectors w_t & w_{t+1}.

PLA update step: $w_{t+1} \leftarrow w_t + y_n x_n$

$$\begin{bmatrix} w_{t+1} \end{bmatrix} = \begin{bmatrix} -1 \\ 0.7 \\ 0.5 \end{bmatrix} + -1 \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2.3 \\ -0.5 \end{bmatrix}$$



c. For the data given, write the equations of hard margin support vector classifier (optimization function and the constraints specific to the data given). Identify how many support vectors exist in the data.

objective : minimize $\frac{1}{2} w^T w$

subject to $y_i (w^T x_i + b) \geq 1$

Constraints:

1. $w_1 + w_2 + b \geq 1$
2. $3w_2 + b \geq 1$
3. $-6w_1 - b \geq 1$
4. $-5w_1 + (-4w_2) - b \geq 1$
5. $2w_1 + w_2 + b \geq 1$
6. $-7w_1 - 9w_2 - b \geq 1$

most likely S.V.

6 Consider the function

$$f(x) = (x+4)^2.$$

Starting at $x_0 = 3$ and using a learning rate of $\eta = 0.01$, compute the value of x after the first iteration of gradient descent.

(a) What is the gradient at $x_0 = 3$? (4 marks)

(b) What is the updated value of x after one step? (6 marks)

Solution: The gradient of f is

$$f'(x) = \frac{d}{dx}(x+4)^2 = 2(x+4).$$

At $x_0 = 3$,

$$f'(x_0) = 2(3+4) = 14.$$

The gradient descent update rule is

$$x_{k+1} = x_k - \eta f'(x_k).$$

Thus, after one step,

$$x_1 = x_0 - \eta f'(x_0) = 3 - 0.01 \cdot 14 = 3 - 0.14 = 2.86.$$

So the updated value is $x_1 = 2.86$.

7 Suppose you have a dataset with 1000 samples.

(a) In batch gradient descent, how many samples are used to compute the gradient at each iteration? (2 marks)

(b) In mini-batch gradient descent with a batch size of 50, how many samples are used per iteration? (2 marks)

(c) If you perform 10 iterations of mini-batch gradient descent (with replacement allowed between iterations), how many total sample usages occur across all iterations? (2 marks)

Solution:

(a) In batch gradient descent, the gradient is computed using the entire dataset, so 1000 samples are used per iteration.

(b) In mini-batch gradient descent with batch size 50, the gradient at each iteration is computed using 50 samples.

(c) Over 10 iterations of mini-batch gradient descent, the total number of sample usages is

$$10 \times 50 = 500.$$

8 Consider a binary classification problem with logistic regression hypothesis:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Given parameter vector

$$\theta = \begin{bmatrix} 0.5 \\ -0.25 \end{bmatrix} \quad \text{and input vector} \quad x = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

and the true label $y = 1$,

(a) Compute $h_\theta(x)$. (3 marks)

(b) Calculate the logistic regression cost for this example. (4 marks)

Solution:

(a) Compute the linear combination:

$$\theta^T x = 0.5 \times 1 + (-0.25) \times 2 = 0.5 - 0.5 = 0$$

Calculate

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5.$$

(b) Calculate the cost:

$$J(\theta) = -1 \times \log(0.5) - 0 \times \log(1 - 0.5) = -\log(0.5) \approx 0.6931.$$

9 Consider a binary classification problem with feature $x \in \mathbb{R}$ modeled by Gaussian Discriminant Analysis.

The class-conditional densities are:

$$p(x \mid y=0) = \mathcal{N}(1,1), \quad p(x \mid y=1) = \mathcal{N}(3,1)$$

and the class priors are:

$$P(y=1) = 0.4, \quad P(y=0) = 0.6.$$

Given an observed feature value $x = 2$,

(a) Compute the likelihoods $p(x \mid y=0)$ and $p(x \mid y=1)$. (4 marks)

(b) Compute the posterior probability $P(y=1 \mid x=2)$. (4 marks)

Solution:

The Gaussian density for scalar x is:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

(a) Calculate the likelihoods:

$$p(x=2 \mid y=0) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2-1)^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\right).$$

$$p(x=2 \mid y=1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2-3)^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\right).$$

Numerically,

$$\frac{1}{\sqrt{2\pi}} \approx 0.3989, \quad \exp(-0.5) \approx 0.6065,$$

so

$$p(x=2 \mid y=0) \approx 0.2420, \quad p(x=2 \mid y=1) \approx 0.2420.$$

(b) Compute the posterior using Bayes' theorem:

$$P(y=1 \mid x=2) = \frac{p(x=2 \mid y=1)P(y=1)}{p(x=2 \mid y=0)P(y=0) + p(x=2 \mid y=1)P(y=1)}.$$

Substitute the values:

$$P(y=1 \mid x=2) = \frac{0.2420 \times 0.4}{0.2420 \times 0.6 + 0.2420 \times 0.4} = \frac{0.0968}{0.1452 + 0.0968} = \frac{0.0968}{0.2420} = 0.4.$$

10 Given a dataset with two features: Weather (Sunny, Rainy) and Temperature (Hot, Mild), and a binary class label: Play Tennis (Yes or No). Assume the Naive Bayes classifier assumption that features are conditionally independent given the class.

Weather	Temperature	Play Tennis
Sunny	Hot	No
Sunny	Hot	No
Rainy	Mild	Yes
Sunny	Mild	Yes
Rainy	Hot	Yes

(a) Explain why Naive Bayes assumes independence between Weather and Temperature given Play Tennis, and discuss one limitation of this assumption. (3 marks)

(b) Calculate the posterior probabilities

$$P(\text{Play Tennis} = \text{Yes} \mid \text{Weather} = \text{Sunny}, \text{Temperature} = \text{Mild})$$

and

$$P(\text{Play Tennis} = \text{No} \mid \text{Weather} = \text{Sunny}, \text{Temperature} = \text{Mild}).$$

Which class is predicted? (4 marks)

Solution:

(a) Naive Bayes assumes conditional independence of features given the class label to simplify calculation of joint likelihood as the product of individual likelihoods. This assumption may not hold in reality, as features could be correlated, which can lead to less accurate probability estimates.

(b) Calculate prior probabilities:

$$P(\text{Yes}) = \frac{3}{5} = 0.6, \quad P(\text{No}) = \frac{2}{5} = 0.4.$$

Calculate likelihoods:

$$P(\text{Weather} = \text{Sunny} \mid \text{Yes}) = \frac{1}{3} \approx 0.333, \quad P(\text{Weather} = \text{Sunny} \mid \text{No}) = \frac{2}{2} = 1,$$

$$P(\text{Temperature} = \text{Mild} \mid \text{Yes}) = \frac{2}{3} \approx 0.667, \quad P(\text{Temperature} = \text{Mild} \mid \text{No}) = 0.$$

Compute (unnormalized) posteriors:

$$P(\text{Yes} \mid \text{Sunny, Mild}) \propto 0.6 \times 0.333 \times 0.667 = 0.133,$$

$$P(\text{No} \mid \text{Sunny, Mild}) \propto 0.4 \times 1 \times 0 = 0.$$

Normalize:

$$P(\text{Yes} \mid \text{Sunny, Mild}) = 1, \quad P(\text{No} \mid \text{Sunny, Mild}) = 0.$$

Prediction: Play Tennis = Yes.

11 (a) Explain the main conceptual difference between bagging and boosting in ensemble learning, focusing on how the base models are trained and combined. (4 marks)

(b) Describe one advantage and one limitation of each method (bagging and boosting). (4 marks (1 each for advantage and disadvantage))

Solution:

(a) **Conceptual difference:** Bagging (Bootstrap Aggregation) trains multiple base models independently and in parallel on different random samples (with replacement) of the training data and combines their predictions by majority vote or averaging. It primarily reduces model variance and helps avoid overfitting.

Boosting trains base models sequentially, where each new model focuses on correcting the errors of the previous models by giving more importance to misclassified examples. The final prediction is a weighted combination of all models. Boosting aims to reduce bias and build strong predictive models from weak learners.

(b) **Advantages and limitations:**

Bagging:
- Advantage: Reduces variance, increasing model stability and reducing overfitting.
- Limitation: May not reduce bias and might underperform if base learners are weak.

Boosting:
- Advantage: Can reduce bias and effectively combine weak learners into a strong model.
- Limitation: More prone to overfitting and sensitive to noisy data and outliers.

You are given the following 3D dataset with 10 points: (20 marks)

[2 x 10=20]

$$X \in \mathbb{R}^{10\times 3} = \begin{pmatrix} 2.5 & 2.4 & 1.0 \\ 0.5 & 0.7 & 0.3 \\ 2.2 & 2.9 & 0.9 \\ 1.9 & 2.2 & 0.8 \\ 3.1 & 3.0 & 1.1 \\ 2.3 & 2.7 & 1.0 \\ 2.0 & 1.6 & 0.5 \\ 1.0 & 1.1 & 0.2 \\ 1.5 & 1.6 & 0.4 \\ 1.1 & 0.9 & 0.3 \end{pmatrix}.$$

1. Create the feature matrix $X \in \mathbb{R}^{10\times 3}$.
2. Compute the mean and standard deviation of each of the three features over the 10 points.
3. Construct the standardized matrix X_{std} in which each feature has zero mean and unit variance.
4. Compute the 3×3 covariance matrix of X_{std}.
5. Compute the eigenvalues and eigenvectors of this covariance matrix.
6. Sort the eigenvalues in descending order and reorder the eigenvectors accordingly.
7. Report all three eigenvalues and the corresponding normalized eigenvectors (principal directions).
8. Form the projection matrix $W \in \mathbb{R}^{3\times 2}$ from the first two principal components.
9. Compute the 2D representation $Z = X_{\text{std}} W \in \mathbb{R}^{10\times 2}$.
10. Create a 2D scatter plot of the rows of Z.